Project Documentation: Global Olympic Sports Performance Analysis

Step-by-step project development procedure

Project Name	Global Olympic Sports Performance Analysis
Tool Used	Tableau (Desktop / Public)
Dataset	athlete_events.csv

1. Introduction

The Olympic Games bring together athletes from all over the world, and every Games creates a large amount of data. This project uses Tableau to study that data and turn it into simple charts, a dashboard and a story. It helps us understand how medals are distributed, which countries take part the most, and which sports are the most popular.

2. Objective

To analyze Olympic Games data and understand:

- medal distribution (Gold, Silver and Bronze)
- participation by country (NOC code)
- participation across different sports

3. Tools and Technologies

Tool	Use
Tableau Desktop / Tableau Public	Data connection, charts, dashboard and story
CSV file	Data source
VS Code and Python (Flask)	Optional website to show the project
GitHub	Storing project files and documentation

4. Dataset Description

The dataset file is **athlete_events.csv** (based on 120 years of Olympic history). Each row is one athlete's entry in one event.

Column	Meaning
ID	Athlete identifier
Name	Athlete name
Sex	Athlete's sex
Age	Athlete's age
Height / Weight	Athlete's height and weight
Team	Team name
NOC	National Olympic Committee code (country code)
Games	Olympic Games edition
Year	Year of the Games
Season	Summer or Winter
Sport	Sport category
Event	Specific event
Medal	Gold, Silver, Bronze, or empty (no medal)

The exact columns may differ slightly depending on the dataset version.

5. Step-by-Step Development Procedure

Step 1: Download the dataset

1. Open a browser and search for the Olympic athlete events dataset.
2. Download **athlete_events.csv** from a legitimate source such as Kaggle.
3. Extract the ZIP file if needed and save the CSV in an easy-to-find folder.

Step 2: Connect the dataset to Tableau

1. Open Tableau.
2. Under **Connect**, choose **Text File**.
3. Browse to **athlete_events.csv** and click **Open**.

Step 3: Check and prepare the data

1. Check that the column names are correct.
2. Check the data types of fields such as Year, Age and ID.
3. Check that Medal, NOC and Sport are available.
4. Look for missing values, especially in the Medal field.
5. Do not delete rows without medals, because they are useful for participation analysis. For the medal chart only, filter Medal to Gold, Silver and Bronze.

Step 4: Create Visualization 1, Medal Analysis

1. Open a new worksheet and rename it **Medal Analysis**.
2. Drag **Medal** to Columns.
3. Drag the record count (Number of Records) to Rows.
4. Filter Medal to Gold, Silver and Bronze.
5. Choose a bar chart and turn on data labels.
6. Add the title **Medal Analysis**.

This chart compares medal records. It counts records, not unique athletes.

Step 5: Create Visualization 2, Olympic Participation by Country

1. Open a new worksheet and rename it **Olympic Participation by Country**.
2. Drag **NOC** to Rows.
3. Drag the record count to Columns.
4. Choose a horizontal bar chart and sort it from high to low.
5. Filter to show only the leading countries so the chart stays readable.
6. Add the title **Olympic Participation by Country**.

NOC is a committee code, not always a country name. The chart counts participation records, not unique people.

Step 6: Create Visualization 3, Sports Participation

1. Open a new worksheet and rename it **Sports Participation**.
2. Drag **Sport** to Rows.
3. Drag the record count to Columns.
4. Choose a bar chart and sort it from high to low.
5. If the list is long, filter to a manageable number of sports.
6. Add the title **Sports Participation**.

Step 7: Build the dashboard

1. Click **New Dashboard** and rename it **Olympic Performance Dashboard**.
2. Choose a suitable size (automatic or fixed desktop size).
3. Drag the three worksheets into the dashboard.
4. Arrange and resize them so every chart is clear.
5. Add the dashboard title **Global Olympic Sports Performance Analysis**.
6. Add filters if needed (for example Year, Season or Medal) and test them.
7. Save the workbook.

Step 8: Create the story

1. Click **New Story** and rename it **Olympic Performance Story**.
2. Drag the dashboard into the story canvas.
3. Add a caption, for example: "This dashboard presents an overview of Olympic medal records, participation across NOC codes, and participation across different sports."

4. Preview the story and save the workbook again.

A dashboard puts many charts on one screen. A story presents views step by step, like slides.

Step 9: Save the workbook

1. Go to **File, Save As**.
2. Save as **Global_Olympic_Sports_Performance_Analysis.twbx** (Packaged Workbook).
3. Keep a backup of the original CSV file.

Step 10: Test the project

Check the following and note what you actually tested:

- All three visualizations appear correctly.
- The medal chart shows Gold, Silver and Bronze.
- The country and sports charts are readable.
- The charts fit inside the dashboard and filters work.
- The story opens and shows the dashboard.
- The saved workbook reopens without errors.

Step 11: Publish and share

1. Create a free account on Tableau Public.
2. Open the workbook and choose **Server, Tableau Public, Save to Tableau Public**.
3. Sign in, name the workbook and save.
4. Check that the dashboard and story open, then copy the link and test it in an incognito window.

Everything published on Tableau Public is public, so do not publish private data.

Step 12: Record the demo video

1. Open the dashboard and start a screen recording.
2. Introduce the title and objective, and explain the dataset.
3. Show and explain each of the three visualizations.
4. Show the complete dashboard and the story.
5. End with the conclusion, stop the recording and save the video.

6. Upload it to the platform your college accepts, set the access permission, and test the link.

Step 13: Upload to GitHub

1. Create a folder named **Global-Olympic-Sports-Performance-Analysis** with the README, workbook, screenshots, demo link and documentation.
2. Create a new repository on GitHub with the same name and a short description.
3. Upload the files with **Add file, Upload files** and commit the changes.
4. Check that all files and links work.

Step 14: Website using Flask (optional)

1. Install Python and Flask (`python -m pip install flask`).
2. Create the folder **olympic-flask-app** with **app.py** and a **templates** folder containing **index.html**.
3. Add the project title, description and the Tableau Public embed code to **index.html**.
4. Run `python app.py` and open `http://127.0.0.1:5000` to check the website.

6. Results and Findings

Add these from your own charts:

- **Medal Analysis:** [which medal has the highest count and the numbers]
- **Participation by Country:** [top 3 NOC codes and their numbers]
- **Sports Participation:** [top 3 sports and their numbers]

7. Conclusion

This project shows how Tableau can turn a large Olympic dataset into clear and useful visuals. With three visualizations, one dashboard and one story, it is easy to understand medal distribution, country participation and sports popularity.

8. Future Scope

- Show country names instead of NOC codes.
- Add year-wise trends for medals and participation.

- Compare Summer and Winter Games.
- Add more filters, such as sex, age group or sport.

9. Limitations

- The charts count records (athlete-event entries), not unique athletes.
- NOC codes may not match current country names.
- The analysis depends on the dataset version used.